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Genome-Wide Association Studies of Delay Discounting and Impulsive Personality Traits in Children From the Adolescent Behavior and Cognitive Development Study

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ABSTRACT

Impulsivity, often operationalized as delay discounting (DD) and as impulsive personality traits via the UPPS-P scales, is a key transdiagnostic construct across psychiatric disorders. Recent genome-wide association studies (GWAS) have studied the genetic basis of impulsivity in adults, but it remains unclear how similar the genetic architecture of DD is in children. The present study conducted GWAS of DD and impulsivity traits in 5548 children (ages 9–10 years old) of genetically inferred European ancestry from the Adolescent Brain Cognitive Development (ABCD) Study. Heritability estimates for DD ($h^2 = 0.20$, S.E. = 0.10) and UPPS-P subscales ($h^2 = 0.08$–0.11 S.E. = 0.05) were comparable to adult estimates. Genetic correlations between adult and child impulsivity were modest ($r_g = 0.28$–0.46), with positive urgency showing the strongest correlation ($r_g = 0.83$). While no genome-wide significant associations were identified, the top associated variants were mapped to genes previously linked to smoking initiation (rs3820908; $p = 6.5 \times 10^{-8}$) and UPPS-P Lack of Premeditation (rs17292179; $p = 4.2 \times 10^{-7}$). Polygenic score (PGS) associations were used to compare the genetic signals in children with those reported in adults. Adult PGSs for DD and positive and negative urgency indicators explained small but significant variance in the respective child impulsivity phenotypes (0.36%–0.44%, $p < 7.5 \times 10^{-4}$). Additionally, UPPS-P indices were broadly associated with PGSs derived from adult externalizing (0.42%–1.02%) and ADHD (0.23%–0.79%). This first GWAS of impulsivity in children offers a developmentally informed comparison of genetic influences, revealing both similarities and differences by developmental stage.

1 | Introduction

Impulsivity, broadly defined as lack of self-control, is associated with a host of adverse health outcomes [1–3] and may function as intermediate mechanisms, or endophenotype, through which genetic variation confers risk [4–6]. However, impulsivity is not a unitary trait [7, 8] and several specific measures of impulsivity are particularly promising endophenotypes for understanding substance use disorders (SUD), behavioral addictions, and other psychiatric disorders [4, 9–10]. Of a number of options, some of the most studied are delay discounting (DD), referring to immediate reward preference [11], and the five-factor UPPS-P impulsive behavior scales [12], referring to several distinct dispositions. Indeed, steeper DD and higher UPPS-P impulsivity scores [13] have been shown to precede the onset of SUDs, with prospective studies indicating that impulsive traits in childhood and adolescence predict later substance use initiation, escalation, and addiction risk [3, 14–15]. Unlike task-based DD measures, the UPPS-P scale [13] measures in five self-reported impulsive personality subdomains—Negative Urgency (e.g., propensity to act rashly during negative mood states), Positive Urgency (e.g., acting out during positive mood states), Lack of Premeditation (e.g., acting without thinking about the consequences), Lack of Perseverance (e.g., difficulty staying focused or giving up easily when challenged), and Sensation Seeking (e.g., the tendency to seek out new, exciting, or risky experiences)—providing a more nuanced framework for assessing different facets of impulsivity. These facets are distinctively associated with various aspects of substance use, with Sensation Seeking and Negative Urgency particularly implicated in the age of initiation [16, 17] and frequency of use [18]. Twin studies have demonstrated DD [19, 20] and facets of impulsive personality scales [21, 22] are heritable in adolescents (~19%–56%) and adults (20%–64%), making them valuable markers for dissecting genetic studies of SUD and other behavioral problems. These phenotypes also have transdiagnostic significance, showing both positive and negative correlations with several psychiatric disorders [4, 9, 23–26].

Recent studies on the genetic basis of substance use [27–35], multiple forms of impulsivity [36–40], and more broadly self-regulation [41] have made significant strides, particularly through large-scale genome-wide association studies (GWAS) in the adult population. SNP-based heritability estimates for measures of impulsivity range from 5% to 10%, confirming the additive genetic component that influences these traits. Several genome-wide significant findings for DD and UPPS-P revealed a broad influence on other behavioral traits [25, 40, 42], many of which are relevant to SUD [43]. Notably, there was also overlap in the significant genetic signals between measures of impulsivity; for example, genetic variants in the *CADM2* were significantly associated with the Sensation Seeking subscale of the UPPS-P model, as well as non-planning and motor from the Barratt Impulsive Scale version 11 [44], and drug experimentation [39].

Advances in the genetics of impulsivity in adults highlight the complexity and importance of these traits as transdiagnostic markers [24, 25]. Building on these findings, there is a pressing need to explore how these genetic components manifest in children, particularly during critical developmental stages. Phenotypic characterization of DD and the UPPS-P have been studied in children and adolescents [45–49], establishing that impulsivity is not only present and multi-faceted but also highly malleable [50–52]. Persistent and elevated impulsivity are often embedded within a broader externalizing spectrum, which includes behaviors such as rule-breaking, aggression, substance use, and psychiatric conditions, such as Attention-Deficit/Hyperactivity Disorder (ADHD). Externalizing traits may share developmental pathways with impulsivity and contribute similarly to a range of psychiatric and behavioral outcomes [53–55]. Understanding these traits during this critical cognitive developmental period offers valuable insights into the trajectory of impulsivity and its impact on behavior with limited substance exposure. Thus, a genome-wide investigation of impulsivity in children offers several key benefits and addresses important gaps in our knowledge. First, it can provide insights into the polygenic contribution to impulsivity in childhood and allow comparisons with adults to assess developmental stability or change in genetic influences [56]. Second, similar to other complex traits, GWAS in children may contribute to a better understanding of the genetic architecture through the identified signals. For example, genetic variants may exhibit higher penetrance [57–59], meaning that the likelihood of expressing the trait in the presence of an associated allele is higher in younger individuals. Finally, given the strong connection between impulsivity and brain development [50, 60], it is essential to understand how genetic variations affect both the tendency for impulsive decision-making and the structural and functional characteristics of the brain.

The Adolescent Brain Cognitive Development (ABCD) study [61] presents an opportunity to simultaneously examine behavioral, neuropsychological, and psychosocial measures in relation to genetic variants in a cohort of 11,875 youths of diverse backgrounds. Our primary aim was to identify genetic variants associated with rigorously QCed DD and UPPS-P traits in children. Our secondary aim was to estimate the SNP-based heritability and genetic correlation between impulsivity indicators in children and contrast them with those derived from external GWASs based on adult populations. Building on these, we sought to validate the associations between impulsivity indicators in children and polygenic scores (PGS) of impulsivity and related phenotypes derived from adult GWASs.

2 | Material and Methods

2.1 | Study Population

The Adolescent Brain Cognitive Development (ABCD) study is an ongoing longitudinal study of brain development and behavioral health that is following more than 11,000 children in the United States for up to 10 years [61]. The study recruited children aged between 9 and 10 years old at baseline from sampled schools from 21 sites across the United State [62]. Written informed consent was obtained from parents or guardians. A wide range of data, including demographics, mental and physical health assessments from each follow-up, multi-omics, and brain imaging, are available [63–65]. This investigation utilized data from release 5.0, focusing on the full cohort, the updated genomic data, and phenotypic measurements in the largest subset of 5548 children of genetically inferred European ancestry based on reference populations.

2.2 | DD Phenotype Assessment

DD was assessed during the first follow-up visit (1-year after the baseline visit) using a computer-based adjusting-amount delay discounting procedure adapted for children [66, 67]. Participating children were presented with 42 hypothetical monetary choices (7-item that is administered 6 times), where they had to decide between receiving a smaller immediate reward (of adjusting magnitude) or a larger reward of $100 after different delay intervals (6 h, 1 day, 1 week, 1 month, 3 months, 1 year, and 5 years). Refer to the manual (https://www.millisecond.com/download/library/v6/delaydiscountingtask/) for the approach followed in ABCD. Several studies have raised issues with the quality of the DD phenotype in ABCD [48, 49, 68], highlighting the need for robust quality control (QC) in this sample. A hyperbolic discounting model [11] was used to estimate the discounting rate (i.e., the *k*-value), and the quality of DD was evaluated using model fit metrics. Specifically, different QC thresholds for the hyperbolic regression model fit R^2 were investigated, taking into account the trade-off between data quality and the potential reduction in sample size [69]. Further details on phenotype quality control and data processing can be found in the Supporting Information. The hyperbolic regression model fit R^2 cut-off was chosen to be ≥ 0.5, reflecting more than half of the variance explained. As a sensitivity analysis, we repeated the heritability estimation for DD across different hyperbolic regression model fit R^2 thresholds. In addition to the primary threshold of $R^2 = 0.5$, we also examined $R^2 = 0$ (no removal), 0.1, 0.2, 0.3, 0.4, 0.6, 0.7, 0.8, and 0.9 to assess whether the heritability estimate of DD was influenced by the choice of model fit threshold. Prior to analysis, DD was log10-transformed and winsorized by replacing values beyond four standard deviations with the nearest non-extreme values.

2.3 | UPPS-P Phenotype Assessment

The UPPS-P impulsive personality scale was administered to assess the five subscales of impulsive personality during the baseline collection. A child-friendly version of the UPPS-P was used, with 20 items simplified for age-appropriate understanding [63]. Responses were collected on a 4-point Likert scale ranging from "Not at all like me" to "Very much like me." We then summed up the respective items without further quality control to calculate Negative Urgency, Positive Urgency, Sensation Seeking, Lack of Planning, and Lack of Perseverance, respectively.

2.4 | Genotyping and Quality Control

Genomic DNA was extracted from saliva samples using the SmokeScreen targeted array based on the Affymetrix Axiom platform [70]. For autosomes and the X chromosome, we used the imputed genotypes as part of the most recent data (ABCD Data Releases 5.0; $n = 11{,}666$), which included stringent QCs. Briefly, samples with mismatched genetic and reported sex were removed ($n = 185$), retaining only unrelated individuals (removing $n = 3490$) that were genetically similar to the European superpopulations (removing $n = 2443$) from the 1000 Genomes Project [71] (Figure S1). We focused on participants in the European genetic similarity group because of the: (1) limited statistical power in non-European genetic similarity groups, (2) potential bias for trans-ancestry meta-analysis with unbalanced sample [72], (3) reduced accuracy of European-derived polygenic risk scores in other genetic ancestries [73]. While this approach limits generalizability, it ensures methodological rigor and interpretability of genetic associations. We then applied additional QC steps to retain sample and genetic variants with missing rate < 0.05, genetic variants with minor allele frequency (MAF) > 0.05, and for autosomal SNPs only: Hardy–Weinberg disequilibrium p-value $> 5 \times 10^{-7}$. After quality control, 5548 unrelated, genetically inferred European samples remained in the analysis. Sample and SNP inclusion and exclusion criteria are summarized in Figure S2.

2.5 | Heritability and Genetic Correlation

SNP-based heritability in ABCD was conducted using genome-based restricted maximum likelihood (GREML) via GCTA [74] on individual-level data and alternatively via LDSC [75] using summary-level data. The model adjusted for continuous covariates, including children's age, their genetic sex, and the first 10 genetic principal components (PCs). While GREML is preferable over methods using summary-level data [76], LDSC has the flexibility to handle genetic correlation analyses using external summary-level data. Thus, we computed genetic correlations of impulsivity traits using GWAS summary statistics derived in ABCD via LDSC and compared them to reported genetic correlations from two external GWASs. These include 3860 young adults aged 24 to 266 from the Avon Longitudinal Study of Parents and Children (ALSPAC) study [77] and over 135,000 adults from the general population that provided consent to participate in 23andMe research [38, 40](Table S1). For each impulsivity trait, we further examined pairwise genetic correlation across the three studies using LDSC.

2.6 | Statistical Analyses

Autosomal GWAS was conducted in PLINK2 [78] using a linear regression model to test the association between the genotype of each SNP, coded additively by the number of minor alleles, and the impulsivity phenotype, while accounting for genetic sex, age at the time of data collection, and the first 10 genetic PCs. Association results for SNPs of interest and their nearby variants were visualized using LocusZoom [79, 80]. For Xchr association analyses, we employed a linear regression model using a 3 degree freedom test for Xchr variants [81] that is robust to male genotype coding and X-inactivation, while adjusting for the same set of covariates as the autosomal model.

Gene-level associations for autosome and Xchr were conducted using MAGMA v1.10 [82] using variant-level p-values and the matching European LD reference panel curated from phase 3 of the 1000 Genomes Project [71]. SNPs were mapped to genes based on their genomic location (genome build 38), and the gene-based p-values were computed using the SNP-wise mean model.

Polygenic scores (PGS) were derived using the LDpred2-auto method [83], which infers the Bayesian posterior distributions of effect size and adaptively evaluates hyperparameters without the validation phenotype as input. GWAS summary statistics from prior large-scale studies were curated for DD [40], UPPS-P [39], adventurousness [84], ADHD [85], and the externalizing factor [41] and (Table S2). We removed variants that mapped to multiple genomic locations (i.e., INDELs), SNPs that had inconsistent reference and alternative alleles between the target population (i.e., ABCD) and external GWAS summary statistics. Each PGS was standardized to have mean zero and unit variance. The performances of PGSs were assessed using the difference in adjusted R^2 between regression models with and without the PGS, accounting for age, genetic sex, and the first 10 genetic PCs. We applied a false-discovery rate (FDR) correction to control for multiple hypothesis testing and considered results with an FDR adjusted p-value < 0.05 as significant. We further examined whether trait-specific PGSs overlapped with other adult

GWAS-derived PGSs in explaining trait variance by comparing the adjusted R^2 from the following covariate-adjusted models to a covariate-only model: (1) trait PGS; (2) all PGSs; (3) all PGS excluding externalizing; and additionally, (4) both trait and externalizing PGSs versus a trait PGS only covariate-adjusted model. All statistical analyses were conducted in R version 4.1.0 [86].

3 | Results

3.1 | Sample Characteristics

The demographic characteristics of the ABCD genetic subsample by sex are in Table 1. Empirical distribution of DD and UPPS-P subscales (Figure S3) demonstrated widespread sex differences across the impulsivity phenotypes, with males ($n = 2959$) generally more impulsive than females ($n = 2589$). Phenotypic correlation patterns were similar between ABCD children and other adult populations (Figure S4B), with moderate positive correlations between the two urgency indicators ($r_{range} = 0.48–0.61$) and between Lack of Premeditation and Lack of Perseverance ($r_{range} = 0.41–0.44$). While DD generally exhibited weaker correlations with UPPS-P facets, the only consistent correlations observed for DD across the three studies (ABCD, ALSPAC, and 23andMe) were with Positive Urgency ($r_{range} = 0.11–0.19$) and Sensation Seeking ($r_{range} = -0.10$ to -0.04). As expected, children in ABCD and young adults in ALSPAC both exhibited higher levels of impulsivity across all traits than the general adult population (Table S1).

3.2 | SNP-Based Heritability and Genetic Correlations

The estimated SNP-based heritability via GREML-GCTA was statistically significant for DD ($h^2 = 0.20$, standard error $= 0.10$, $p = 0.04$) and Positive Urgency ($h^2 = 0.11$, S.E. $= 0.05$, $p = 0.04$; Figure 1). Except for Negative Urgency and Sensation Seeking, which did not show significant heritability, the remaining UPPS-P indicators had moderate heritability estimates between

TABLE 1 | Summary of phenotypes and demographic variables stratified by sex subgroups in ABCD children.

Delay Discounting logk	Boys (N = 2959)	Girls Girls (N = 2589)	Overall (N = 5548)	Two-sample *p*-value
Age at Baseline				
Mean (SD)	10.9 (0.764)	10.9 (0.759)	10.9 (0.762)	0.25
Parental income				
< $5 k	26 (0.9%)	21 (0.8%)	47 (0.8%)	0.67
$5 k-11,999	46 (1.6%)	27 (1.0%)	73 (1.3%)	
$12 k-15,999	27 (0.9%)	23 (0.9%)	50 (0.9%)	
$16 k-24,999	67 (2.3%)	68 (2.6%)	135 (2.4%)	
$25 k-34,999	93 (3.1%)	88 (3.4%)	181 (3.3%)	
$35 k-49,999	172 (5.8%)	157 (6.1%)	329 (5.9%)	
$50 k-74,999	388 (13.1%)	349 (13.5%)	737 (13.3%)	
$75 k-99,999	461 (15.6%)	429 (16.6%)	890 (16.0%)	
$100 k-199,999	1111 (37.5%)	935 (36.1%)	2046 (36.9%)	
$200 k+	390 (13.2%)	369 (14.3%)	759 (13.7%)	
Mean (SD)	−2.09 (0.702)	−2.20 (0.704)	−2.14 (0.71)	< 0.001
Negative Urgency				
Mean (SD)	8.61 (2.59)	8.15 (2.56)	8.40 (2.59)	< 0.001
Positive Urgency				
Mean (SD)	7.97 (2.82)	7.47 (2.80)	7.74 (2.82)	< 0.001
Sensation Seeking				
Mean (SD)	10.3 (2.61)	9.61 (2.61)	9.96 (2.63)	< 0.001
Lack of Premeditation				
Mean (SD)	8.13 (2.34)	7.57 (2.27)	7.87 (2.33)	< 0.001
Lack of Perseverance				
Mean (SD)	7.18 (2.18)	6.96 (2.23)	7.08 (2.21)	< 0.001

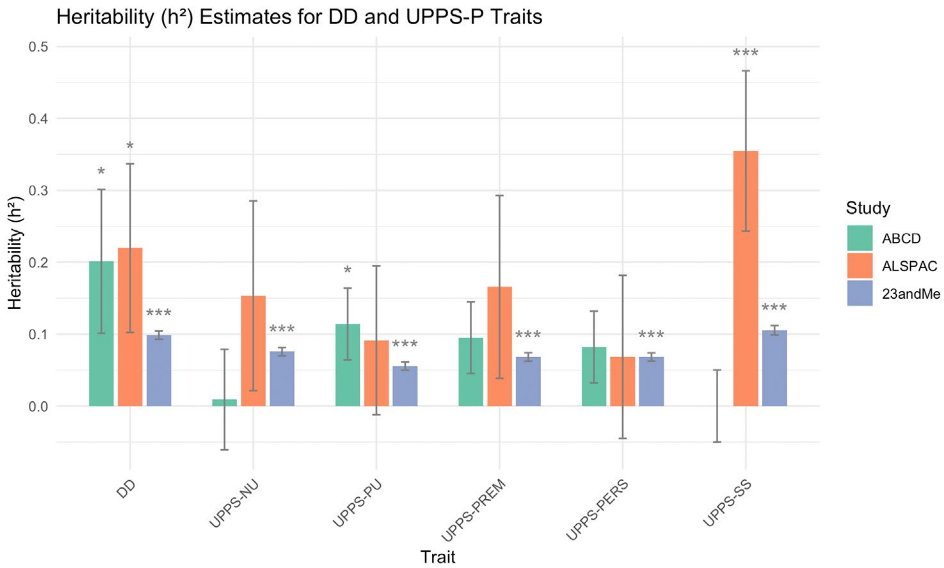

FIGURE 1 | Estimated SNP-based heritability for impulsivity across three population studies. The bar plot summarizes the SNP-based heritability (h^2) estimates for delay discounting and the five UPPS-P impulsivity subscales across three cohorts: ABCD, ALSPAC, and 23andMe. Heritability estimates for ABCD was based on GCTA results, and those for ALSPAC and 23andMe were derived using LDSC. Asterisks indicate statistically significant estimates ($p<0.05$, $p<0.01$, $p<0.001$), with several traits showing significant heritability across multiple cohorts, particularly delay discounting.

TABLE 2 | Estimated SNP-based heritability using GCTA.

Impulsivity trait	Estimate h^2	S.E.	p-value (two-sided)
Delay Discounting	0.201	0.10	0.038
Negative Urgency	0.009	0.07	0.902
Positive Urgency	0.114	0.05	0.038
Sensation Seeking	0.000	0.05	1.000
Lack of Perseverance	0.082	0.05	0.128
Lack of Premeditation	0.095	0.05	0.056

0.082 and 0.0955 (S.E. = 0.05; Figure 1; Table 2). The LDSC estimates were similar to GREML but with much larger standard errors (Table S3) and did not yield any significant results. Across the three studies, we found estimated heritability of DD to be most robustly supported (Figure 1). There was also moderate consistency in Positive Urgency, with significant heritability estimates in both ABCD and 23andMe samples, and in Sensation Seeking, with significant estimates in ALSPAC and 23andMe (Figure 1). For DD, the estimated heritability was not sensitive to different model fit thresholds, with estimates ranging between 0.12 and 0.22 (GCTA) and increasing S.E. as the number of samples being removed at more stringent thresholds (Table S3).

There was one marginally significant LDSC genetic correlation between Lack of Premeditation and Lack of Perseverance ($r_g=0.97$, S.E. = 0.46, $p=0.04$; Table S4) in ABCD. This was also the only consistent finding across studies, similar to the $r_g=1$ reported in ALSPAC and $r_g=0.5$ reported in 23andMe (Figure S4). Specifically, when examining the genetic correlation of the same impulsivity trait across studies, there was stronger agreement between 23andMe and young adults in ALSPAC, with 4 out of 6 measures showing significant correlations ($r_g=0.47$–0.80; Figure S5). In contrast, the genetic correlation between 23andMe and ABCD was significant only for Positive Urgency ($r_g=0.83$, S.E. = 0.34; Table S5).

3.3 | Genome-Wide Association Studies in ABCD

As expected given the modest sample size, the GWAS did not yield any genome-wide significant markers (Tables S6 and S7; Figure S6), with no sign of type I error inflation (Figure S7) and the genomic control lambdas were within an acceptable range (1.001–1.019). There was one signal just below the genome-wide significance: between DD and variants in the *HECW2* gene on chromosome 2 (lead SNP rs3820908; $p=6.5\times10^{-8}$; MAF = 0.29), and three suggestive signals with $p<1\times10^{-7}$: between Negative Urgency and variants in the *DOCK1* gene on chromosome 10 (lead SNP rs59879440; $p=1.26\times10^{-7}$; MAF = 0.1); and between Lack of Premeditation and rs6928101 ($p=2.7\times10^{-7}$; MAF = 0.49) in the *KCNQ5* gene on chromosome 6, and rs17292179 ($p=4.2\times10^{-7}$; MAF = 0.06) near the *CADM2* gene on chromosome 3 (but not in LD with previously reported *CADM2* SNPs). In particular, the *STK17B–HECW2* gene region (Figure S8) contains several genome-wide significant hits (in moderate LD with the lead SNP identified in ABCD data; $R^2=0.36$–0.37) from published GWASs of smoking initiation and brain phenotypes, such as cortical thickness and surface area. The MAGMA analysis did not identify any gene-based association (Table S8).

3.4 | Polygenic Score Analysis

DD and the UPPS-P urgencies PGSs derived using adult GWASs were associated with their corresponding phenotypes in children

(Table S9). Trait-specific PGSs explained 0.36%–0.44% of the phenotypic variance ($p=7.5\times10^{-4}$–4.7×10^{-7}). While the PGS for Lack of Premeditation was not associated with the trait itself, it was significantly associated with both urgency indicators (0.09% and 0.22%, $p=0.016$ and 2.3×10^{-4}) and Lack of Perseverance (0.15%, $p=2.5\times10^{-3}$). The externalizing PGS was broadly associated with all UPPS-P indices (adjusted $R^2=0.42\%$–1.02%; $p=8.5\times10^{-7}$–1.9×10^{-14}), but not with DD (Figure 2). Similarly, the ADHD PGS was also associated with all UPPS-P indices (adjusted $R^2=0.23\%$–0.79%; $p=1.9\times10^{-4}$–1.8×10^{-11}), except Sensation Seeking, which was strongly associated with the adventurousness PGS (adjusted $R^2=0.93\%$; $p=2.6\times10^{-13}$). Except for DD, all impulsivity measures showed improved variance explained when incorporating additional PGSs, with a significant non-overlapping contribution from the externalizing PGS beyond trait-specific predictors (Figure S9).

4 | Discussion

This study represents the first GWAS of impulsivity phenotypes in children. We observed significant heritability for DD and Positive Urgency in children from the ABCD study, supporting the role of genetic influences on these traits early in development. Moreover, for the first time, we confirmed a broad overlap in genetic architecture between child and adult traits for DD and both urgency indicators, suggesting at least partial continuity in the underlying genetics across developmental stages.

While we confirmed partial genetic overlap between children and adult impulsivity for some, however, the remaining impulsivity indicators were not associated with their PGSs. This was surprising as we had successfully replicated these associations in adult populations [7, 77]. In the case of Lack of Premeditation, it was possibly due to the smaller sample size of ABCD children, which could limit the power of the analysis to surpass the FDR correction. In the other case, we hypothesize that a lack of association could reflect the early, undifferentiated nature of these traits, which may separate into more distinct facets of impulsivity with age. For instance, Sensation Seeking was significantly associated with other related PGSs derived in adults (e.g., adventurousness and externalizing) but not the trait PGS itself. Further, it has poor internal consistency as compared to the other UPPS-P indicators in ABCD was also noted [87].

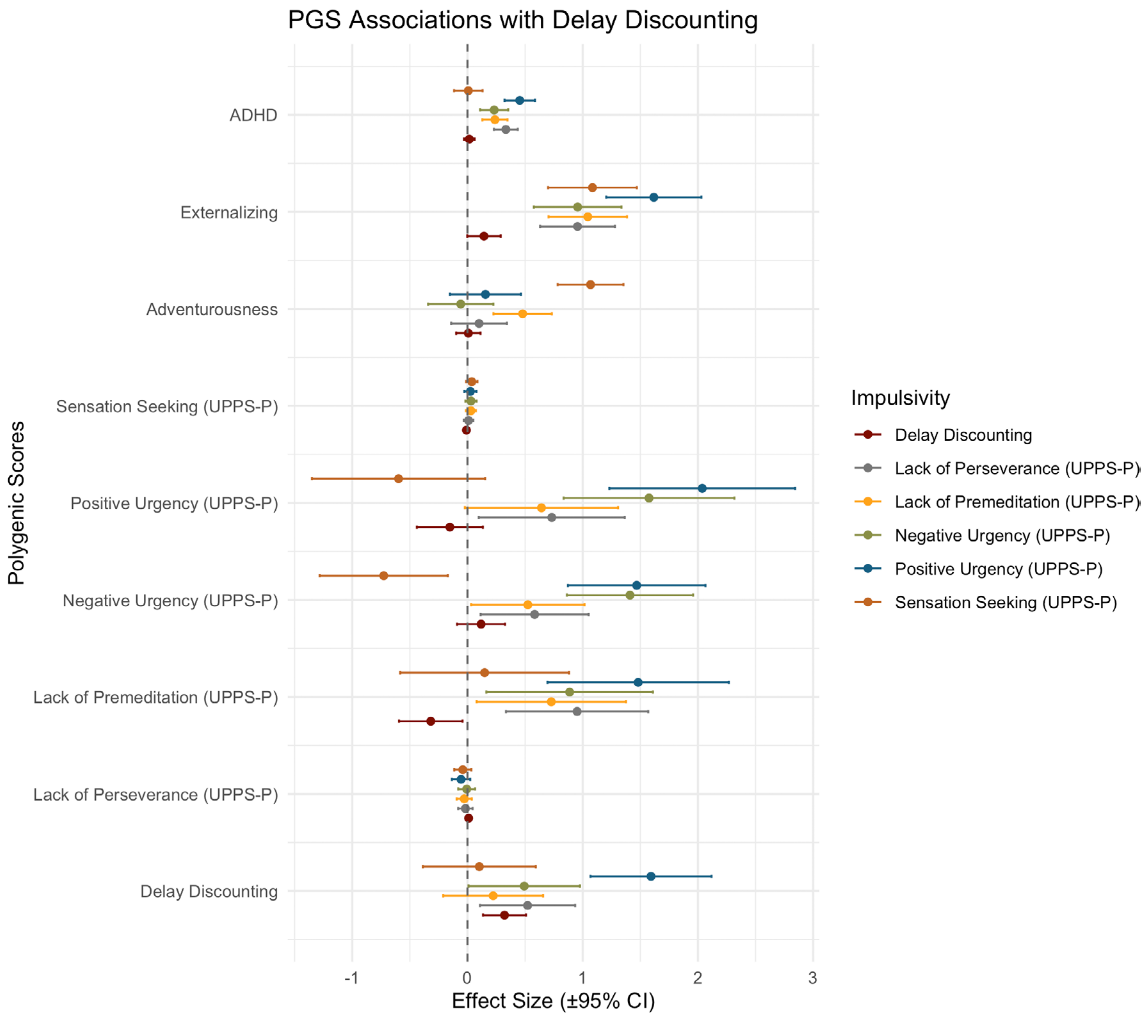

FIGURE 2 | PGSs association with impulsivity traits. The *y*-axis reflects standardized regression coefficients (i.e., effect sizes), and error bars denote 95% confidence intervals. Each point represents the association between a given PGS and an impulsivity phenotype—delay discounting and the five UPPS-P subscales—estimated using linear regression. PGS-impulsivity associations are staggered by trait within each PGS group. The dashed horizontal line indicates the null (zero effect).

More broadly, our results are in line with other studies using adult-derived PGS in the ABCD cohort. We confirmed the associations between externalizing and the majority of impulsivity phenotypes found in another study [88]. Our results also confirmed a similar lack of associations between PGS for externalizing behaviors and DD [89]. Interestingly, the ADHD PGS was also associated with multiple UPPS-P phenotypes measured in children, suggesting different facets of childhood impulsivity tapping into the genetic basis of inattention and hyperactivity that are signatures of ADHD at both early adolescence and adulthood [90], which may become more pronounced over time [91]. These polygenic overlaps and shared liability across behavioral traits were supported by the observation that in addition to trait-specific PGS, related PGSs also contributed significantly to trait variance in ABCD for all UPPS-P indicators (Figure S9). This suggests that while the genetics of related traits and impulsivity overlap, some facets of childhood impulsivity (e.g., Sensation Seeking and Lack of Perseverance) may be less aligned with their corresponding adult traits, potentially reflecting developmental or context-dependent differences in their manifestation.

Finally, DD had a few sub-genome-wide associations with several genetic variants in *HECW2*, but these associations were not observed in *prior* GWASs of DD in adults. The *HECW2* gene has key roles in a number of neurodevelopmental processes, such as the proliferation, migration, and differentiation of neural crest cells that regulate the glial cell line-derived neurotrophic factor (GDNF)/Ret signaling [92, 93]. Rare mutations in this gene have also been robustly linked to neurodevelopmental delay, epilepsy, behavioral abnormalities, and autistic spectrum disorders [94–98], which are also characterized by elevated impulsivity [99–102], but not with DD specifically. The other interesting finding was the replication of variants near the *CADM2* gene with Lack of Premeditation that was previously reported in adults [39].

Several limitations bear mentioning. A primary limitation of the present GWAS is the modest sample size, which limits statistical power for genome-wide discovery, which was further exacerbated by the loss of samples due to the need for aggressive QC of the DD phenotype. While the ABCD cohort is the largest sample available for these traits in childhood, these findings remain underpowered compared with adult GWASs for similar phenotypes. In particular, our results suggest moderate heritability for UPPS-P indicators (6%–10%), similar to values reported in adult populations. To achieve $>80\%$ power to confirm these estimates, a sample size of at least $n=9000$ to 15,000 would be required. Additionally, although the ABCD study is more diverse than many previous large-scale behavioral genetics cohorts, multi-ancestry GWASs still require larger sub-sample sizes to yield ancestry-specific discoveries and ensure generalizability of PGS across populations. Finally, while the psychometric properties of the UPPS-P Impulsive Behavior Scale have been established in ABCD [103], data quality of DD has been problematic in terms of (1) the most appropriate method to generate summary measure of DD [104, 105], (2) low effort/low attention performance can violate assumptions of the test [105].

To conclude, these results reveal, for the first time, the molecular genetic correlates of diverse impulsivity phenotypes in children, pointing to additive genetic influences and difference in genetic architecture compared with adults. Future longitudinal genetic studies, as have recently been undertaken in adults [77], are warranted to determine whether these genetic influences remain stable or change across development, providing a true life-course perspective. Integrating genetic data from children in longitudinal cohorts will permit tracking changes in impulsivity over time, revealing the interplay between genetic and environmental factors during early development when children and their brains are both malleable, but also vulnerable.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Genotype and phenotype data of ABCD Data Release 5.0 are available through the NIMH Data Archive pending a successful application. The PGS analyses included summary statistics from published GWASs made available by the Psychiatric Genomics Consortium (PGC), the UK Biobank, the Externalizing Consortium, and 23andMe Inc. Due to concern for participants' privacy, the full summary statistics of delay discounting and the five UPPS-P subscales are available for academic use upon request from 23andMe. The full summary statistics based on ABCD data will be made available via GWAS catalog (https://www.ebi.ac.uk/gwas/).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.